

BOEKBESPREKINGS / BOOK REVIEWS

Multivariate Analysis *by Anant M. Kshirsagar*

First edition, xiv + 534 pages, Marcell Dekker, Inc., New York, 1972. Price \$19-50.

The author considers the theory of those multivariate distributions underlying some of the multivariate techniques, such as multivariate regression, analysis of variance and covariance, canonical correlations, discriminant analysis and principal components analysis. (Factor analysis is not dealt with.) Thus with these statistical applications in mind there is no opportunity for him to drift into irrelevant distribution theory. Instead the test criteria for the particular analysis are defined and their distributions derived, usually with a discussion on or reference to the particular asymptotic distributions. Applications of the theory are described and references to actual examples in the literature are given. Unfortunately the book lacks a system of presenting the numerous definitions and results which it contains. Amongst the last 400 pages there are only 11 theorems and one definition formally presented as such. With these few exceptions each subsection of the book is written in continuous form without any formal tabulation of results or demarcation of the scope of proofs, a situation which is undesirable in a textbook. To aggravate this, the book is entirely typewritten and therefore lacks the usual typographical shortcuts to clarity such as heavy print, characters of different sizes, etc.

An outstanding feature of the book is an excellent bibliography at the end of each chapter and good referencing within the chapter. Thus postgraduate students with a thorough grounding in vector space algebra, matrix theory and mathematical statistics at the undergraduate level should find this an excellent reference textbook in any study of the theory and applications of multivariate techniques.

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Preservation of Infinite Divisibility under Mixing *by F.W. Steutel*

First edition 99 pp. Mathematisch Centrum, Amsterdam 1970. \$3.00

In this thesis the author studies certain classes of infinitely divisible distributions which, when mixed, remain infinitely divisible. There is a short section on mixtures of exponential distributions which motivates the main classes of distributions to be studied, namely the so-called $\Gamma(\alpha)$ -distributions, with characteristic function $((\lambda/\lambda - it)^\alpha (\alpha > 0))$, and generalization to distributions with c.f. $((\lambda/\lambda - h(t))^\alpha$. Mixing with respect to λ or α or both is considered.

For $\Gamma(\alpha)$ -distributions the cases $0 < \alpha \leq 1$, $1 < \alpha \leq 2$, $\alpha > 2$ are distinguished. In the first case it is proved that all mixtures of $\Gamma(\alpha)$ -distributions ($0 < \alpha \leq 1$) are infinitely divisible. For $\alpha > 1$ examples are given of mixtures which are not infinitely divisible. It is proved, however, that the set $\{\alpha \mid \int_0^\infty (1-itx)^{-\alpha} dG(x) \text{ is inf. div.}\}$ has a maximum α_0 such that $1 \leq \alpha_0 \leq 2$. The author conjectures that $\alpha_0 = 2$ and supplies some evidence in favour of his conjecture.

The generalization has not been treated extensively, but nevertheless a couple of very interesting and powerful results have been obtained. The thesis is concluded with a chapter on related infinitely divisible distributions where topics from renewal theory, monotone densities, lattice distributions, etc. are discussed.

One of the basic techniques of this work is to prove the result for finite mixtures and then to apply a suitable limit theorem.

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